

RespiPredict

Multimodal Respiratory Disease
Prediction, Analysis and Forecasting System



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Abstract

RespiPredict is a cross-platform AI-powered system for early respiratory disease prediction, analysis, and forecasting. It accepts four types of patient input — chest X-rays, breathing audio, EHR/Excel records, and a patient questionnaire — and combines them through a multimodal machine learning fusion engine to generate personalised risk scores, vulnerability assessments, and future health forecasts.

Multimodal Input

4 data types fused for accuracy

AI-Powered Forecast

Predicts future risk trends

Personalised Output

Unique analysis per patient

Introduction

The problem RespiPredict addresses



Respiratory diseases (COPD, asthma, TB, pneumonia) are among the leading causes of global death, affecting hundreds of millions annually.



Diagnosis is often delayed because patient data — imaging, vitals, history, audio — is fragmented across systems and never analysed together.



No accessible home-based tool exists for individuals to monitor respiratory risk proactively and receive personalised forecasts.



RespiPredict bridges this gap: one AI system that unifies all inputs and delivers early-warning risk insights to patients and clinicians.



Existing System

Limitations of current approaches

Current Tools

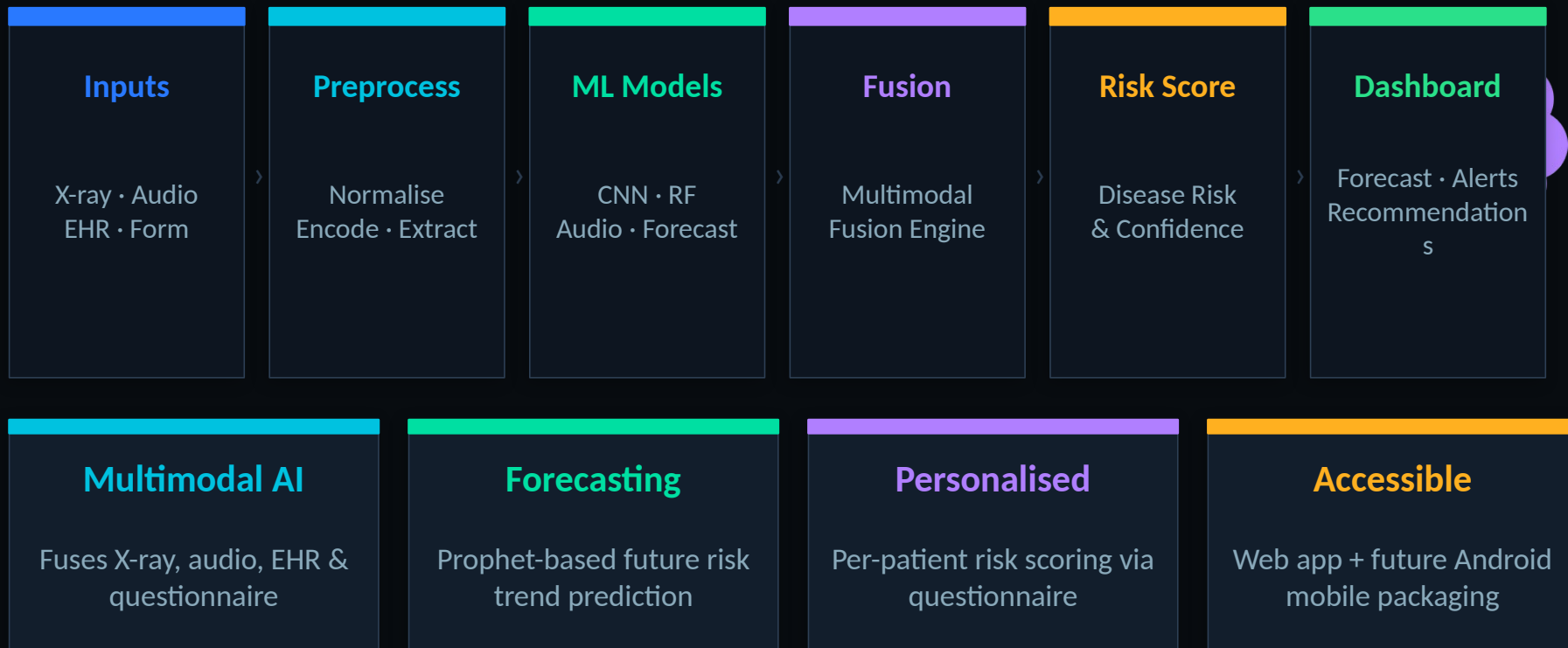
- Single-modality tools (X-ray OR EHR only)
- Hospital PACS / radiology software
- Standalone spirometry devices
- Generic health apps without AI
- Manual clinical assessments only

Limitations

-  No multimodal data fusion
-  No future risk forecasting
-  Not accessible for home use
-  No personalised risk profiling
-  Cannot combine AQI + vitals + audio

Proposed System

RespiPredict — the solution



System Architecture

End-to-end data flow through RespiPredict



Chest X-Ray

JPEG/DICOM



**Breathing
Audio**

WAV/MP3



EHR / Excel

1,500 records



Patient Form

Questionnaire

↓ Preprocessing & Feature Extraction ↓

CNN Module

X-ray classification

Audio Module

MFCC classifier

EHR ML Module

Random Forest

Form Module

Risk encoding

↓ Multimodal Fusion Engine ↓



Fusion Model combines all 4 model outputs into one unified risk score + confidence



Risk Score + Forecast + Alerts

Dashboard output to user

Features of the Project



Chest X-Ray Classification

CNN / Transfer Learning classifies X-rays (Pneumonia, TB, COVID-19, Lung Opacity) using PyTorch + torchvision



Breathing Audio Analysis

MFCC & spectrogram features extracted via librosa for acoustic respiratory condition classification



EHR / Excel Risk Prediction

Random Forest stacking classifier on 1,500-row synthetic dataset with 25+ patient features



Patient Questionnaire Engine

Personalised risk from lifestyle, symptoms, vitals, AQI location, and full medical history inputs



Future Risk Forecasting

Prophet time-series forecasts respiratory vulnerability over 30 / 90 days from historical inputs



Multimodal Fusion Engine

All four specialist model outputs fused into one unified risk score and confidence level for the patient

Datasets Used

Data sources that power RespiPredict

01

Chest X-Ray Dataset (Kaggle)

~5,800 images

Classes: Normal, Pneumonia (Bacterial/Viral), COVID-19, Tuberculosis, Lung Opacity. Used for CNN transfer-learning classification model.

02

Local X-Ray Dataset

Custom split

Subset used for smoke-testing the CNN pipeline. Classes: Normal, Lung Opacity, Viral Pneumonia. Validated preprocessing flow.

03

Synthetic EHR Dataset

1,500 records · 25+ columns

Generated for development. Features: Age, SpO2, Resp. Rate, AQI, Smoking, BMI, Symptoms, Medical History, Disease Risk, Future Risk score.

04

Respiratory Audio Dataset

Pending — ICBHI 2017 planned

Local audio files inspected and found silent. ICBHI 2017 Respiratory Sound Database (920 recordings) identified as valid replacement for MFCC training.

System Requirements



Hardware

Processor : Intel Core i5+ / AMD Ryzen 5+

RAM : 8 GB minimum (16 GB recommended)

Storage : 10 GB free disk space

GPU : NVIDIA (optional, for CNN training)

Internet : Required for AQI API & Colab



Software

OS : Windows 10/11 or Ubuntu 20+

Language : Python 3.9+

Framework : Streamlit · PyTorch · scikit-learn

Libraries : librosa · Plotly · pandas · NumPy

Tools : VS Code · Google Colab · Figma

Module Description

Part 1 — Input, Preprocessing, X-Ray & EHR/Audio

Input Module



1

Accepts four data types via the Streamlit UI: chest X-ray images (JPEG/DICOM), WAV/MP3 audio samples, EHR Excel uploads, and a patient health questionnaire form.

Preprocessing Module



2

Images resized to 224×224 and normalised. Audio processed through librosa for MFCC & spectrogram extraction. Tabular data cleaned, label-encoded, and standard-scaled.

X-Ray CNN Module



3

Transfer-learning CNN (ResNet-50 / EfficientNet) fine-tuned on Kaggle chest X-ray dataset. Classifies into: Normal, Pneumonia, COVID-19, Tuberculosis, Lung Opacity.

EHR / Audio Module



4

Random Forest stacking classifier trained on 1,500-row synthetic EHR dataset (25+ features). Parallel audio CNN/ML classifier uses MFCC features to detect breathing anomalies.

Module Description

Part 2 — Fusion, Forecasting & Output Dashboard



5

Multimodal Fusion Module

Aggregates prediction outputs from the CNN, EHR/audio, and questionnaire models. A meta-learner (stacking or weighted averaging) produces one unified risk score (0–100) with a confidence percentage. This is the core intelligence of RespiPredict.



6

Forecasting Module

Uses Facebook Prophet (time-series forecasting) to project how the patient's respiratory risk score is likely to evolve over the next 30 and 90 days, based on historical data, seasonal trends, AQI values, and current vitals.



7

Output / Dashboard Module

Streamlit-based dashboard presents: disease risk prediction with probability bars, a vulnerability badge (Low / Moderate / High), the forecast chart, contributing factor breakdown, automated health reminders, and personalised next-step recommendations.

Output Screen

What RespiPredict displays to the user

Dashboard



Risk score overview, AQI status, and latest vitals summary at a glance.

Assessment Results



Disease prediction list with probability scores and model confidence rating.

Risk Forecast



30/90-day respiratory risk trend chart generated by the Prophet model.

Vulnerability Alert



Risk badge (Low / Moderate / High) with top contributing health factors.

Recommendations



Personalised action plan: doctor visit, lifestyle changes, re-assessment date.

X-Ray Analysis



Uploaded chest X-ray displayed with the predicted disease class and confidence.

Future Enhancement

Planned improvements and extensions

Near-term

Complete CNN training on full Kaggle X-ray dataset

Integrate valid ICBHI 2017 respiratory audio dataset

Add SHAP / Grad-CAM model explainability

Mid-term

Android APK packaging (Kivy / BeeWare)

Home-screen widgets and push notifications

Doctor-facing patient PDF report export

Long-term

Real-time AQI API integration via GPS location

Wearable device input (SpO2 smartwatch data)

Clinical validation and accuracy benchmarking

Conclusion

RespiPredict demonstrates how multimodal AI can address real-world healthcare challenges. By fusing chest X-rays, breathing audio, EHR records, and patient questionnaire data through a unified ML engine, it delivers accurate disease risk prediction, vulnerability analysis, time-series forecasting, and personalised health guidance — all accessible through a clean, intuitive web interface.

Multimodal

4 input types unified

Predictive

ML-based risk
scoring

Forecasting

Future trend
analysis

Actionable

Personalised
guidance

Accessible

Web + Mobile ready



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Thank You

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Questions?

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For AI-assisted risk prediction only — not a substitute for professional medical diagnosis.